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SECTION 02

Project Proposal

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Problem Statement

Many Mandarin learners struggle to correctly write and recall Chinese characters. They frequently forget the stroke order and lose interest as a result of traditional memorisation techniques that are repetitive and boring. This slows down their learning and frustrates them. Therefore, there is a need for an interactive and interesting learning system that can lead users step-by-step, provide them immediate feedback on their writing accuracy and eventually improve their ability to remember characters.

AI Solution

Rydit: Learn Mandarin is an AI-driven mobile application that transforms traditional Mandarin writing practice into an interactive and personalized learning experience. The app uses artificial intelligence to **analyse user handwriting** in real time, by recognising the stroke order, directions and overall accuracy. The app uses AI handwriting recognition to compare the user's input with the correct stroke sequences and immediately provides visual feedback, guiding the learner to improve precision and consistency in writing Chinese characters.

Apart from handwriting recognition, the AI also powers the app's **adaptive learning engine**. It tracks each user's performance every time they answer a question and dynamically adjusts lesson difficulty based on their strengths and weaknesses. Using the Leitner spaced repetition method [1], Rydit ensures that characters and words the user struggles with will reappear more frequently, while mastered ones are reviewed once in a while to strengthen memory retention. The lessons are challenging but rewarding to engage the users in an immersive learning experience, while learning how to write in mandarin painlessly.

Additionally, Rydit gamifies the entire learning process with progressive levels, rewards and interactive challenges to sustain motivation. The AI monitors engagement patterns and suggests personalised practice sessions to prevent boredom or burnout. By combining adaptive AI feedback, smart repetition and game-based learning, Rydit offers a fun, efficient, and scientifically grounded way for learners to master Mandarin writing and vocabulary.

Goal of AI Solution

Rydit: Learn Mandarin aims to give learners a more efficient, customized and entertaining method of learning Chinese character writing. By teaching users step-by-step and providing real-time feedback on stroke order, direction and structure, the AI seeks to increase writing accuracy [2]. Additionally, AI adjusts the complexity of lessons according to user performance, which boosts motivation and learning at a personal pace [3]. Through customized guidance and on-going assessment, research indicates that personalized AI systems can improve retention, fortify self-regulated learning habits and maintain learners' increased engagement [4][5]. By using interactive challenges, adaptive learning and real-time feedback, the AI seeks to help users write more precisely, retain characters longer and maintain the motivation throughout their Mandarin learning process.

Process of Empathize in Design Thinking

Empathy Map

SAYS ● "How to write the characters correctly? " ● "I can recognize the words but always forgot how to write." ● "I always forget the stroke order." ● "It's so hard to learn Mandarin."	**THINKS** ● "I wish there was a fun way to learn!" ● "I want to learn at my own pace." ● "It would be great if there is an app to help me in learning." ● "Traditional memorization is boring."
DOES ● Find a tuition teacher. ● Watch YouTube tutorials on writing characters. ● Practice on a paper or tablet. ● Seek help from friends. ● Learn from app like Duolingo	**FEELS** ● Frustrated when forgetting how to write. ● Confused about stroke order. ● Excited when recognizing or writing correctly. ● Satisfied when seeing progress or getting feedback
PAINS ● Forgetting too fast and lack of motivation to practice. ● Hard to memorize characters without writing repeatedly. ● Forgetting old characters easily. ● Boring if learning alone.	**GAINS** ● A system that help them to remember in long term ● Step-by-step writing animations for better understanding. ● AI feedback on stroke order and accuracy. ● Fun and interactive experience that keeps them motivated.

Process of Define in Design Thinking

User:

- An individual who is learning Mandarin language.

Need:

- The user needs a way to remember and recall Chinese characters. The user forgets learned characters fast.
- The user needs a way to practice writing correctly and confidently. The user has identified a pain point associated with the challenge of writing the characters accurately and remembering the stroke order.
- The user expressed a demand for an interesting way of learning the language. The users think that learning Chinese can be boring.
- The users think that they would be more motivated to learn with others but at the same time learning at their own pace.

Insights:

- Retention. Spaced repetition is vital to ensure that learned words are not forgotten.
- Writing Accuracy. Writing practice is also important to ensure the characters are written correctly.
- Motivation. Forgetting characters causes frustration and reduces motivation.
- Engagement. Users would enjoy learning in a gamified way and learning in an interactive way.

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